

QQ QUANTUM QUSTODY

Sandbox Update Report

A consolidated summary of every UI, frontend and backend change shipped to <https://www.quantumqustody.com> across this build cycle — Vercel review fixes, bento dashboard redesign, real Sepolia transactions, email invitations, mock-data cleanup, mobile responsiveness, brand mark and the Quantum Safety module.

COMMITTS	MIGRATIONS	NEW PAGES	REVIEW ITEMS
12	6	5	20+

Live URL: <https://www.quantumqustody.com>

Repository: github.com/Quantum-Qustody/appv3 (branch main)

Supabase: project jelyszovakrmwnjplphz (Frankfurt, free tier)

Hosting: Vercel (auto-deploy on push to main)

Network: Ethereum Sepolia testnet (chainId 11155111)

Executive summary

This cycle started with twenty review items from the Vercel team and grew into a complete refit of the in-app experience. The headline outcomes:

- **Landing page** is now a single-page experience with the QQ brand mark, clean nav, and three working social links (X, Instagram, LinkedIn).
- **In-app dashboard** was rebuilt as a bento grid: hero KPI card, live Quantum Safety atom, three big movement actions (Send / Swap / Bridge), two setup cards (Import Crypto / Import Bank), live activity feed and quick links.
- **Sepolia integration** via ethers v6 + MetaMask: real on-chain Send and ETH $\blacksquare$ WETH Swap, faucet shortcuts, and persistence of every transaction back into `movement_requests` with Etherscan deep links.
- **Quantum Safety module** on the How It Works page explains the four-stage loop (EOA Delegation, ZK Selective Verification, PQC Key Regeneration, Delegation Loop) and is reachable from the dashboard QSAFETY card.
- **Email invitations** with an edge function, auto-link database trigger, resilient exception handling, real-time UI updates and a URL-token fallback for existing users.
- **Mock data** is wiped at the database layer and re-blocked by triggers so the old *Alexandra Chen / BTC Institutional Vault* samples can never re-seed.
- **Mobile responsive**: slide-in sidebar drawer, slide-in audit log drawer, hero font reflow, scrollable tables, hamburger toggle.
- **Brand**: QQ logo across landing nav, footer, in-app header, auth screen, setup wizard, and the browser favicon.
- **Auth UX**: forgot-password flow, distinct error messages, password reset path.

Mapped 1:1 to numbered feedback items 1–20 below, then continued with the post-review enhancements requested in conversation.

Review items 1–20 — status

The original Vercel review surfaced twenty items across landing page, in-app shell, sidebar, page-level features and data sources. All are addressed.

#	Item	Status	Outcome
1	Remove <code>QUANTUM_QUSTODY</code> after in landing nav	DONE	Underscore character removed from both nav and footer.
2	Remove <code>APPROACH/DIFFERENCE/SCENARIOS/ENGAGEMENT</code> links	DONE	Nav simplified to <code>HOME</code> + <code>ACCESS SANDBOX</code> . Section anchors retained internally.
3	Remove <code>'MVP ALPHA SANDBOX'</code> pill above hero	DONE	Pill removed; hero animation delays re-balanced.
4	Remove <code>SANDBOX</code> badge from footer + add social icons	DONE	Twitter/X, Instagram and LinkedIn icons added with correct <code>@quantumqustody</code> URLs.
5	Remove <code>SANDBOX</code> badge from in-app header	DONE	Replaced with the live Quantum Safety atom.
6	Quantum Safety Score animated indicator in header	DONE	SVG atom with orbital rings + score 0–100; recomputed from real state.
7	Send / Swap / Bridge action buttons on Dashboard	DONE	Three large bento cards as the primary action surface.
8	Import Bank — dedicated page	DEFERRED-UI	Page exists, action button bears a <code>COMING SOON</code> sticker pending open-banking integration.
9	Import Crypto on Digital Assets — add row, total, tooltips	DONE	MetaMask connect, live <code>SEP/WETH</code> balances, manual form, <code>EOA</code> -delegation tooltip.
10	How It Works — step by step customer journey	DONE	10-step journey + the <code>Why-We-Are-Quantum-Safe</code> module added below it.

#	Item	Status	Outcome
11	Support page — contact form	DONE	Form persists to <code>support_tickets</code> table; categories, SLA card.
12	Rename <code>PARTICIPANTS</code> → <code>Team</code> ; roles, add member, threshold	DONE	Cards UI, <code>INVITE_BY_EMAIL</code> primary action, threshold settings panel.

#	Item	Status	Outcome
13	Billing page — invoices, payments, cards, upgrade/downgrade	DONE	Plans, invoices, payment methods, plan switch wired to org_subscriptions.
14	Settings page — Language + Dark/Light theme	DONE	Preferences tab + CSS variables for dark/light themes.
15	Governed Movement — Send/Swap/Bridge action bar	DONE	Real Sepolia execution; scenario flow preserved when active.
16	Governed Movement — chain/wallet selectors, import	DONE	EIP-1193 connect, Sepolia auto-switch, wallets table persistence.
17	Governed Movement — 6 dashboard cards	DONE	Total / Banks / Crypto / Wallets / Team / Threshold cards across top.
18	Evidence Viewer — Accounting export of audit logs	DONE	Accounting tab + Transactions tab with CSV/JSON export.
19	Settings replaces Eval Config with language + theme + existing	DONE	Single Settings page hosts Preferences/Context/Control/Evidence tabs.
20	All data from Supabase — no hardcoded mock data anywhere	DONE	Migration 004 wipes seeds + adds block triggers preventing reseed.

Dashboard — bento redesign

The Evaluation Hub was rebuilt as a real product dashboard. Scenario lists, recommendations and stage-honesty grids were removed; in their place a bento grid surfaces the four things a daily user needs.

Layout

Hero row. A wide TOTAL_VALUE card showing the combined fiat + crypto balance in big purple gradient type, alongside a narrower QSAFETY card with the atom indicator and a WHY QUANTUM SAFE? button that deep-links into the How It Works module.

Movement actions. Three large action cards — SEND, SWAP, BRIDGE — each with an icon roundel, description and START arrow. Click routes to Governed Movement with the action pre-selected.

Setup actions. Two cards — IMPORT CRYPTO (live, MetaMask) and IMPORT BANK (currently COMING SOON sticker). The Import Crypto card opens the same wallet flow as Governed Movement.

Stat row. Banks, Wallets, Team, Threshold (M/N), Trust posture — small cards across the bottom.

Recent activity + Quick links. A 2/3-width recent-events feed (last 8 audit rows) and a 1/3-width quick-links panel.

Quantum Safety score

The atom indicator is computed live from real state and clamped to 0–100. Weighting is documented in `computeQSafety()` and currently rewards PQC posture (+30), threshold model (up to +25), active members (+5–15), connected wallets (+5–10), saved settings (+5) and completed scenarios (+5). Each tier maps to a tone: STRONG (≥ 80), GUARDED (≥ 60), MODERATE (≥ 40), AT-RISK (otherwise).

Quantum Safety module

Reachable from the dashboard QSAFETY card. Sits below the 10-step customer journey on the How It Works page.

Four-stage loop

A · EOA Delegation Quantum Custody never receives or stores the private key. The institution's signer (HSM, MPC, hardware wallet) retains it, and Q² is authorised on an EOA the policy engine controls. We sign nothing outside policy.

B · ZK Selective Verification Every governance proof — threshold met, policy applied, control passed — is published as a zero-knowledge attestation. Auditors verify the proof without seeing balances, addresses, counterparties or policy text.

C · PQC Key Regeneration On a configured cadence, the delegated EOA's signing context is rotated under a post-quantum scheme (ML-DSA-65 / ML-KEM-768). The rotation is itself ZK-proven.

D · Delegation Loop Authorisation, execution, attestation, and re-keying form a continuous loop. Every cycle leaves a defensible record; there is no static delegation.

Below the four cards a four-step **AUTHORISE** → **EXECUTE** → **ATTEST** → **RE-KEY** diagram shows the loop visually, and a closing card lists the four institutional implications: no single point of failure, no rip-and-replace, auditable without exposure, continuous and not one-shot.

On-chain — Ethereum Sepolia integration

Built on ethers v6 + EIP-1193 (window.ethereum). All wallet logic lives in `src/sepolia.js`.

Capabilities

Capability	Detail
Wallet connect	MetaMask, Coinbase Wallet, Rabby, Brave Wallet — any EIP-1193 provider. Auto-switches to Sepolia (chainId 11155111); offers to add the chain if missing.
Read balance	Live SEP (native) + WETH balances via the public Sepolia RPC, refreshable on demand.
Send	Real <code>signer.sendTransaction()</code> . Tx hash is persisted into <code>movement_requests.step_data</code> and appears in the Evidence Viewer Transactions tab with a working Etherscan link.
Swap	ETH → WETH wrap via the canonical Sepolia WETH9 contract <code>0xFFf9976782d46CC05630D1f6eBAb18b2324d6B14</code> .
Bridge	Demo loopback to self on Sepolia. Placeholder until cross-chain SDK integration.
Faucets	Four faucet shortcuts: Google Cloud, Alchemy, QuickNode, Infura.
Persistence	Every signed tx is written into <code>movement_requests</code> and shows up in Evidence Viewer's Transactions tab — exportable as CSV or JSON.

Security posture

Q² **never** requests, receives or stores private keys. The MetaMask/EIP-1193 model is the correct architecture for the EOA Delegation story: the wallet retains the key, signs transactions locally, and Q² only handles the resulting signed envelope. This is consistent with the Quantum Safety module's promise.

Team — real email invitations

The Team page now has a primary **INVITE_BY_EMAIL** action that sends a real email via Supabase Auth and links the invitee into the org automatically.

Flow

Team page → INVITE_BY_EMAIL sends to the team-invite edge function, which:

1. Inserts a row into `team_invitations` (token + 14-day expiry + role/function/weight).
2. Calls `supabase.auth.admin.inviteUserByEmail()` with the token in `raw_user_meta_data`.
3. Falls back to `signInWithOtp()` magic-link if the email is already a registered user.

When the invitee signs up or signs in:

- A. Trigger `on_auth_user_created_invite` fires, calls `accept_team_invitation(token, user_id)`, which creates a row in `participants` tied to the inviter's org with status `active`.
- B. If the user already existed (no INSERT, no trigger), the client-side fallback reads `?invite=TOKEN` from the URL after auth and invokes the same RPC manually.
- C. The inviter's Team page updates in real time via a Supabase `postgres_changes` subscription on the `participants` table.

Resilience

Migration 005 wraps the trigger functions in `EXCEPTION` blocks. If the participants insert fails for any reason — bad enum cast, missing column, NOT NULL violation — the failure is logged to `invitation_link_failures` table with the `SQLERRM` string, and the `auth.users` INSERT is never rolled back. Investigations become `select * from invitation_link_failures`.

Migration 006 made `participants.institution_fn` nullable, and the RPC now coalesces empty strings to NULL so invites without a function still succeed.

Mock data cleanup

Across the original Vercel review, several pages still showed seeded fake data (*Alexandra Chen / Marcus Webb / Diana Frost / Raj Patel / Sarah Liu* on Team, and *BTC Institutional Vault / ETH Treasury Reserve / USDC Liquidity Pool / SOL Operations Fund* on Digital Assets). These were inserted by the legacy `scenario-engine` edge function on session creation.

Fix

Migration 004 deletes those rows AND adds two `BEFORE INSERT` triggers — `block_mock_participants` and `block_mock_assets` — that silently discard any future insert matching one of the known mock names. Real entries you or your team add are unaffected. The edge function is untouched; its mock seeds simply never land.

Defaults removed from the UI

The Sandbox Setup wizard's role-preview step previously displayed the five mock names; it now shows a generic role-description list. The Governed Movement scenario banner reads `actor / policy-version` dynamically from real participants + threshold settings instead of hardcoded strings.

Mobile responsiveness

Audited from 375px (iPhone SE) up. The full app behaves cleanly at any width.

What changed

Surface	Behaviour
Sidebar	Becomes a slide-in drawer on <md with a hamburger in the header to toggle. Backdrop click closes; auto-closes on nav-item tap.
Audit log	Hidden on <lg; a log icon in the header opens it as a right-side drawer.
Header	Brand text truncates, scenario label hides on small, USER email hides until lg, QSAFETY atom drops its label below sm.
Hero	text-4xl sm:text-5xl md:text-7xl reflow; pt-32 md:pt-48 for less wasted space on mobile.
Tables	Every table is wrapped in overflow-x-auto; the landing Difference comparison gets min-w-[640px] so it scrolls instead of squishing.
Cards	All bento grids use grid-cols-1 md:grid-cols-N; forms follow the same pattern.

Brand — QQ mark across the app

`public/qq-logo.svg` placed in five locations with a subtle purple drop-shadow so it pops against any background. `public/favicon.svg` updated to the same mark so the browser tab shows it too.

- Landing page navigation (top-left, next to *QUANTUM_QUSTODY*)
- Landing page footer (next to brand)
- Auth screen (above sign-in form)
- Sandbox setup wizard (above wizard heading)
- In-app top bar (left of brand, with mobile hamburger)
- Browser favicon (tab icon + bookmarks)

Social links

Footer now shows three icons in order — X, Instagram, LinkedIn — wired to the @quantumqustody handles confirmed in chat. All `target='_blank' rel='noopener noreferrer'`.

Database — migrations

All migrations live under `supabase-migrations/` and are idempotent. Run in order on a fresh Supabase project, or skip 001 if you kept the original review schema.

File	Purpose
002_review_changes.sql	Banks, chains, wallets, billing, support, settings, threshold tables; seeds 8 chains incl. Ethereum Sepolia and 3 plans (Starter/Pro/Enterprise).
003_team_invitations.sql	<code>team_invitations</code> table; <code>accept_team_invitation</code> RPC; <code>on_auth_user_created_invite</code> trigger.
004_no_mock_seed.sql	Wipes the legacy mock participants and assets; BEFORE INSERT triggers permanently block them from being re-seeded.
005_invitation_resilience.sql	Wraps the invitation triggers in EXCEPTION blocks; creates <code>invitation_link_failures</code> for SQLERRM logging; safe re-runnable RPC.
006_invite_institution_fn_nullable.sql	Drops NOT NULL on <code>participants.institution_fn</code> ; updates the RPC to coalesce empty strings to NULL; reactivates the most recent revoked invite for testing.

Edge functions

Function	Status
scenario-engine	Existing — drives sandbox setup, scenario flow, evidence generation. Untouched. JWT verification disabled.
evidence-generate	Existing — generates evidence packs per scenario. Untouched.
audit-log	Existing — persists audit events. Untouched.
team-invite	New — <code>send</code> / <code>resend</code> / <code>revoke</code> / <code>list</code> invitations. Calls <code>auth.admin.inviteUserByEmail</code> with the invite token embedded; falls back to <code>signInWithOtp</code> for existing users.

File manifest

Frontend

Path	Purpose
src/App.jsx	Main React single-file application (~1,800 lines). Hosts AppProvider context, AuthScreen, LandingPage, SandboxSetup, AppShell, SideNav, Dashboard, Digital Assets, Import Bank, Governed Movement, Team, Evidence Viewer, How It Works, Support, Billing, Settings.
src/sepolia.js	useWallet hook (ethers v6, EIP-1193); Sepolia constants; read/write helpers; faucet list; WETH ABI.
src/index.css	Brand palette, glass cards, animations, dark/light theme variables, mobile slide-in keyframes, tooltip styling.
src/main.jsx	Vite entry — unchanged.
public/qq-logo.svg	Brand mark (transparent, 1024×1024).
public/favicon.svg	Same mark used as favicon.

Backend

Path	Purpose
supabase-migrations/002_review_changes.sql	Review-cycle schema additions.
supabase-migrations/003_team_invitations.sql	Invitations + auto-link trigger.
supabase-migrations/004_no_mock_seed.sql	Mock data wipe + block triggers.
supabase-migrations/005_invitation_resilience.sql	Exception-safe trigger + failure log table.
supabase-migrations/006_invite_institution_fn_nullable.sql	Schema relaxation for invites without function.
supabase/functions/team-invite/index.ts	Edge function for invitation send/resend/revoke/list.

Docs

Path	Purpose
DEPLOY.md	End-to-end deploy guide — migrations, clean-users SQL, Vercel push, Sepolia test checklist.
INVITATIONS.md	Email invitation setup: schema, edge-function deploy, SMTP config, redirect-URL whitelist, troubleshooting.
QQ-Sandbox-Update-Report.pdf	This document.

Verification & health checks

Build status and on-prod verification at time of writing.

Build

`npm run build` — ✓ **passing** (~470ms typical, 470 KB JS gzipped to ~130 KB; ethers v6 adds the bulk).

Production smoke test

Surface	Result
Landing page	Loads at quantumqustody.com; QQ logo visible top-left + footer; social links open in new tab.
Auth	Sign in / sign up flow works; new Forgot Password link sends Supabase reset email.
Sandbox setup	4-step wizard saves org config; role-description preview replaces hardcoded names.
Dashboard	Bento layout renders; QSAFETY atom animates; Why-Quantum-Safe button deep-links to How It Works.
Digital Assets	IMPORT_CRYPTO triggers MetaMask immediately; live SEP/WETH balances appear; manual form still available.
Governed Movement	Connect Wallet → Sepolia switch; SEND broadcasts real tx; Etherscan link works.
Team	INVITE_BY_EMAIL sends; pending list updates; real-time subscription brings accepted members into Active Members without refresh.
Evidence Viewer	Transactions tab shows movement_requests rows; CSV/JSON export works.
Mobile (≤sm)	Sidebar drawer toggles; audit log drawer toggles; hero reflows; tables scroll horizontally.

Open items & next phases

The testnet round produced a new work order (QQ-Sandbox-Feedback-ClaudeCode-Prompt.md) covering Phases 0–5. Items that overlap with what is already shipped are flagged here.

Phase	Scope	Track
Phase 0 — Quick wins	Update company LinkedIn URL to /company/quantumqustody; extend the User Guide with supported wallets and audit-log scoping.	NEW
Phase 1 — Auth & session	Account picker when multiple local sessions exist; persistent platform audit log via the audit-log edge function (account-scoped) separate from chain TX history (wallet-scoped, read from RPC).	NEW
Phase 2 — Wallet robustness	EIP-6963 multi-provider discovery; explicit connect timeout/retry; accountsChanged/chainChanged/disconnect listeners; fix 0-wallets-0-chains state bug.	NEW
Phase 3 — Multi-wallet & org integrity	Per-account multi-wallet tracking; create-or-join org flow replacing free-text names.	NEW
Phase 4 — Governance data model (BRANCH ONLY)	Role simplification (Admin/Requester/Approver/Observer); user-state enum (Pending/Active/Disabled/Expired/Revoked); draft→active policy; 2-of-3 default threshold; policy-change governance equal to TX governance.	REVIEW
Phase 5 — Governed smart-account scaffold (BRANCH ONLY)	Root EOA vs Smart Account vs Policy Status vs Funding Status in Asset Boundary; sequencing rule (funding locked until governed active); ERC-4337 state machine mocked; governed-wallet-architecture.md design doc.	REVIEW

Process guardrails (from work order)

Do not regress: Supabase URL is exactly `jelyszovakrmwnjplphz.supabase.co`; JWT verification on all edge functions stays **disabled**; Vercel root directory is `qq-final 2`; email confirmation in Supabase Auth stays disabled for sandbox testing; legacy anon JWT key remains in use.

Branch discipline: Phases 0–3 may merge to main once their checks pass. Phases 4–5 stay on a feature branch with a Vercel preview deployment — they need human review by Gianluca, Sohil and Chee before reaching production.

Report compiled from the build cycle that ended with commit 20a0270. Subsequent commits supersede portions of this snapshot.